03/02/26 Recitation :- 4

A univ wants to predict whether a student will pass or fail exam based on academic factors.

Study Hours	Attendance	Assignment	Internal Score	Result
12	85	80	35	1
6	60	55	22	0
15	90	88	40	1
5	58	50	20	0
10	75	70	30	1
8	65	60	25	0

Write the sigmoid function used in logistic ~~function~~ regression.

$$z = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 \quad w_0 = b$$

$$a(z) = \frac{1}{1+e^{-z}} \qquad \text{Error} = y - \hat{y}$$

$$\text{loss} = -\left[y \log(\hat{y}) + (1-y) \log(1-\hat{y}) \right]$$

$$w_i^{New} = w_i^{old} + \eta \frac{1}{N} \sum (y - \hat{y}) x_i$$

$$b_i^{New} = b_i^{old} + \eta \frac{1}{N} \sum (y - \hat{y})$$

Iteration 1, epoch 1

$w_1 = w_2 = w_3 = w_4 = 0$

$b = 0$

$\eta = 0.1$

$z = 0$

$\hat{y} = \frac{1}{1+e^{-0}} = 0.5$

$y - \hat{y}$
$1 - 0.5 = 0.5$
$0 - 0.5 = (-0.5)$
$1 - 0.5 = 0.5$
$0 - 0.5 = (-0.5)$
$1 - 0.5 = 0.5$

$\Sigma (y - \hat{y}) x_i$

$= (0.5 \times 12) + (-0.5 \times 6) + (1.5 \times 0.5)$

$+ (-0.5 \times 5) + (0.5 \times 10)$

$= 17$

$$\text{log loss} = -\frac{1}{N} \Sigma \left[y \log(\hat{y}) + (1-y) \log (1-\hat{y}) \right]$$

$y = 1$,

$-\log(0.5) = 0.693$

$y = 0$, $-\log(1-0.5) = -\log(0.5) = 0.693$

$6 \times 0.693 = 4.158$

$\frac{4.158}{6} = 0.693$

$$w_i^{New} = w_i^{old} + \eta \frac{1}{N} \Sigma (y - \hat{y}) x_i$$

$$w_1 = (0.5 \times 12) + (0.5 \times 6) + (0.5 \times 15) + (-0.5 \times 5) + (0.5 \times 10) + (0.5 \times 8)$$

$$= 17$$

$$w_i = 0 + 0.1 (17/6) = 0.283$$

$$w_2 = 1.642$$

$$w_3 = 1.608$$

$$w_4 = 0.733$$

$$b = 0.017$$

$$z = 0.283(6) + 1.642(60) + 1.608(50) + 0.733(22) = 204.78$$

$$\hat{y} = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-204.78}} = 1$$

Error $\therefore$ Epoch 2

$y - \hat{y}$

$1 - 1 = 0$ $\quad$ $1 - 1 = 0$

$0 - 1 = (-1)$

$1 - 1 = 0$

$0 - 1 = (-1)$

$1 - 1 = \square$

$\hat{y} = 1, y = 1$

$-\log(1) = 0$

$\hat{y} = 1, y = 0$

$-\log(1-1) = -\log(0) = \infty$

$-\log(1-0.999) = 6.9$

$6.9 + 6.9 = 13.8$

log loss $\frac{13.8}{6} = 2.3$

$\therefore$ Epoch 3

$w_1 = 0.1, w_2 = (-0.325), w_3 = (-0.142),$

$w_4 = (0.033)$ $b = -0.016$

$N = 6, \eta = 0.1$

$z = -36.65$

$\hat{y} = \frac{1}{1+e^{+36.65}} = 0$

Error

$1-0 = 1$

$0-0 = 0$

$1-0 = 1$

$0-0 = 0$

$1-0 = 1$

$1-0 = 1$

$y = 1, -\log(0.001) = 6.907$

$4(6.907) = 27.628$

$y = 0$

$-\log(1-0.001) = 0.001$

$2(0.001) = 0.002$

$$\frac{27.628 + 0.002}{6} = \boxed{4.605}$$

$\therefore$ Epoch 1, 0.693

Epoch 2, 2.3

Epoch 3, 4.605